

Summary

We propose a new loss function to reconstruct images from measurements only, i.e., **without any ground truth example**. It exploits the invariance to transformations of usual image distributions and equivariant network architectures to **recover the missing information**.

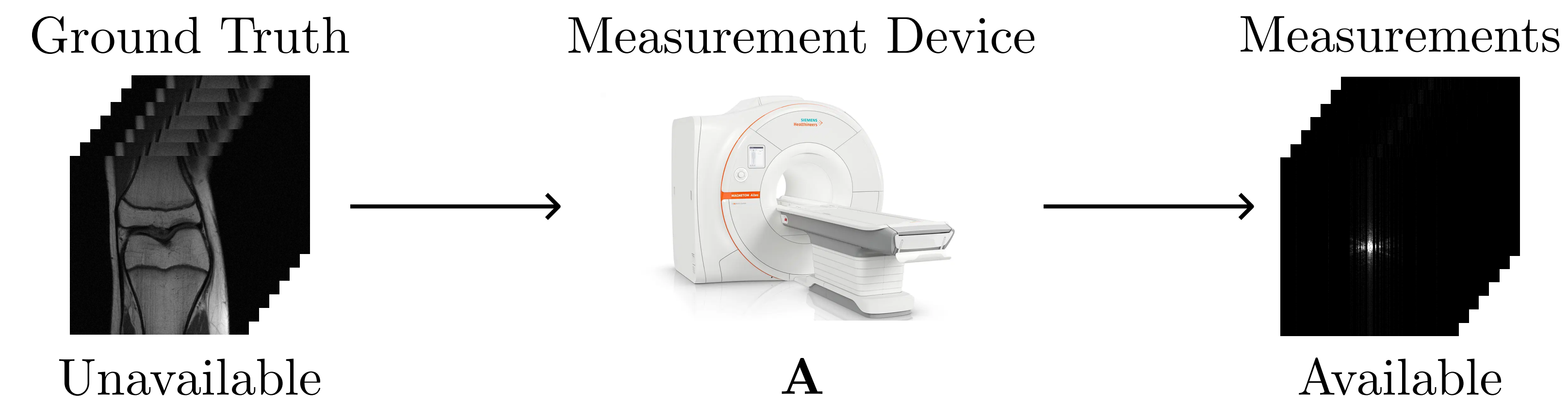
Self-supervised learning for inverse problems

Problem setup Let $x \in \mathcal{X} \subsetneq \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_m)$

$$y = Ax + \varepsilon$$

Goal find $f(y, A) \approx x$

Challenge use only $(y_i)_{i \in I}$



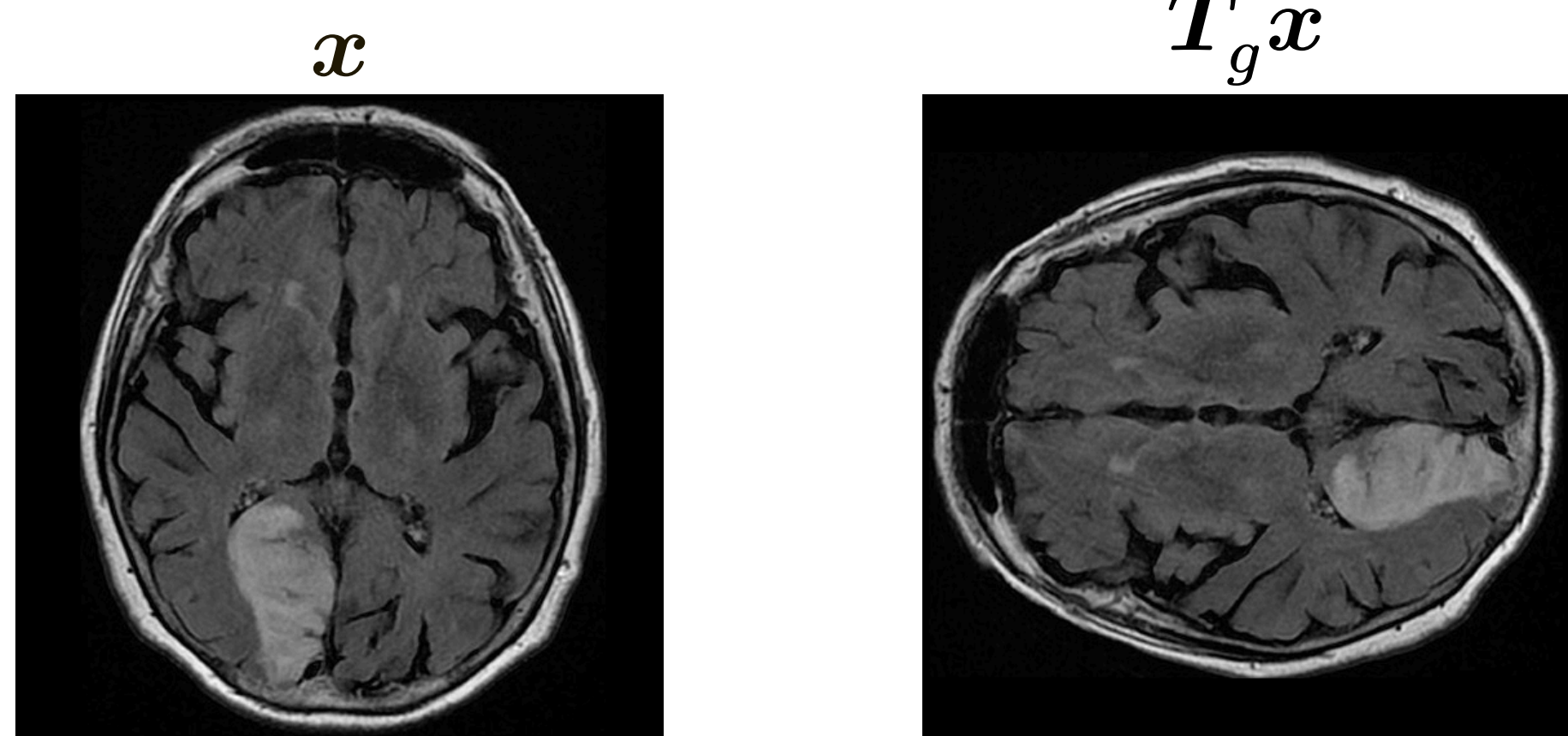
Key ingredient: invariances of image distributions

Assumption invariance of $p(x)$ to transformations

$$p(T_g x) = p(x) \quad \forall x, g$$

Virtual operators

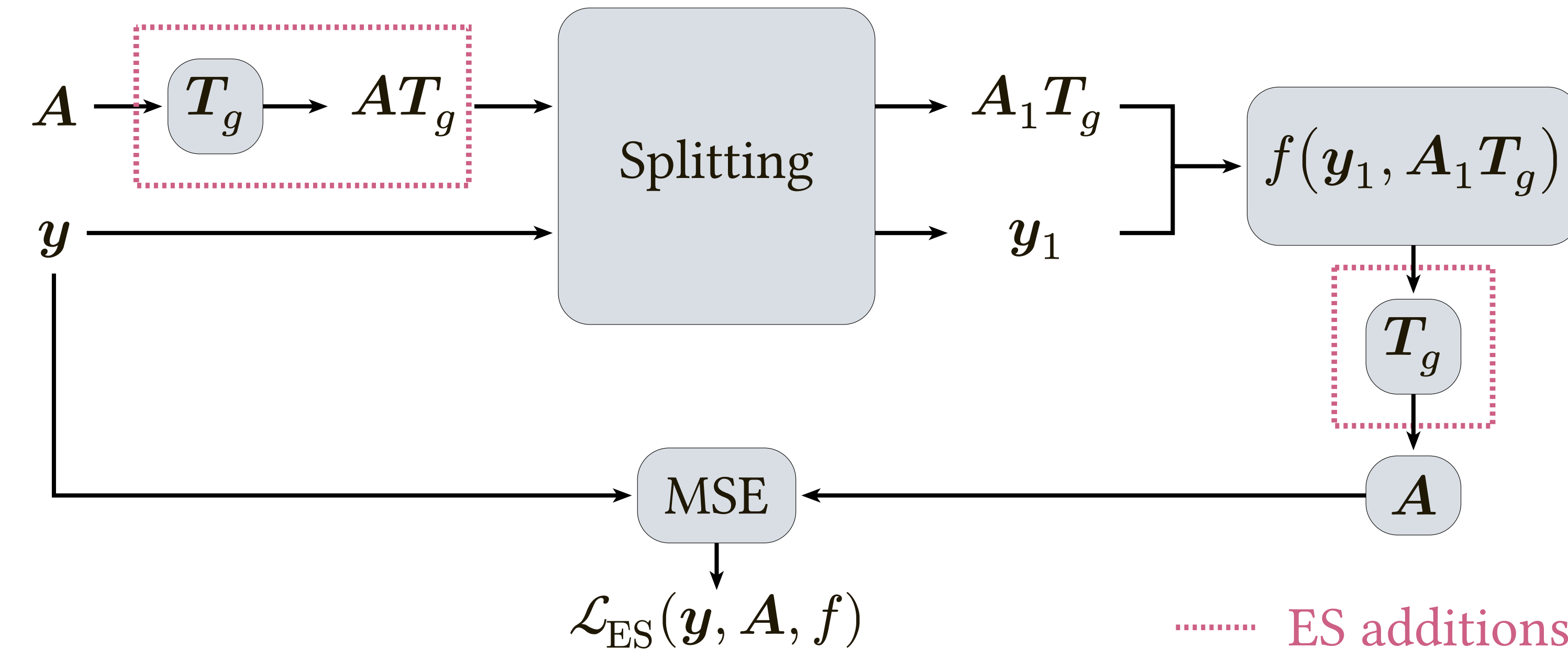
$$y = Ax = AT_g \underbrace{T_g^{-1}x}_{x' \in \mathcal{X}} = A_g x'$$



Equivariant splitting

$$\text{Splitting: } y = (y_1^\top, y_2^\top)^\top = (A_1^\top, A_2^\top)^\top x$$

$$\mathcal{L}_{\text{ES}}(y, A, f) \triangleq \mathbb{E}_{y_1, A_1 | y, AT_g} \left\{ \|AT_g f(y_1, A_1 T_g) - y\|^2 \right\}$$



Definition (Equivariant reconstructors)

The reconstructor $f(y, A)$ is equivariant if

$$f(y, AT_g) = T_g^{-1} f(y, A), \quad \forall y, A, g$$

Examples of equivariant reconstructors

- **Reynolds averaging** for any reconstructor $r(y, A)$

$$f(y, A) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} T_g r(y, AT_g)$$

- **Artifact removal** for an equivariant denoiser $\phi(x)$

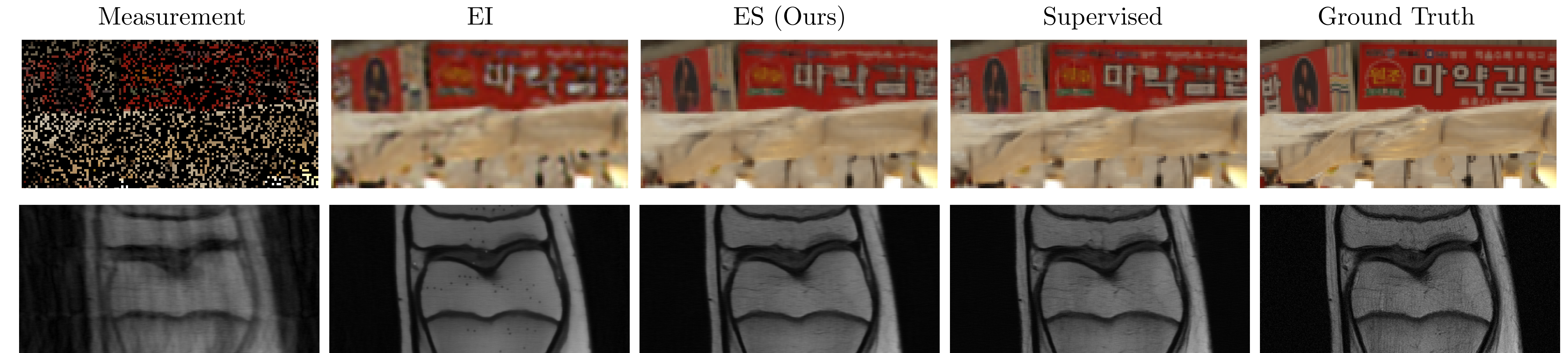
$$f(y, A) = \phi(A^\top y), \quad \text{or} \quad f(y, A) = \phi(A^\dagger y)$$

- **Unrolled networks** for an equivariant denoiser $\phi(x)$

$$\begin{cases} f(y, A) = x_L \\ x_{k+1} = \phi(x_k - \gamma \nabla_{x_k} d(Ax_k, y)) \\ x_0 = 0 \end{cases}$$

- **MAP** and **MMSE** estimators for an invariant distribution $p(x)$

Reconstructions for inpainting and MRI

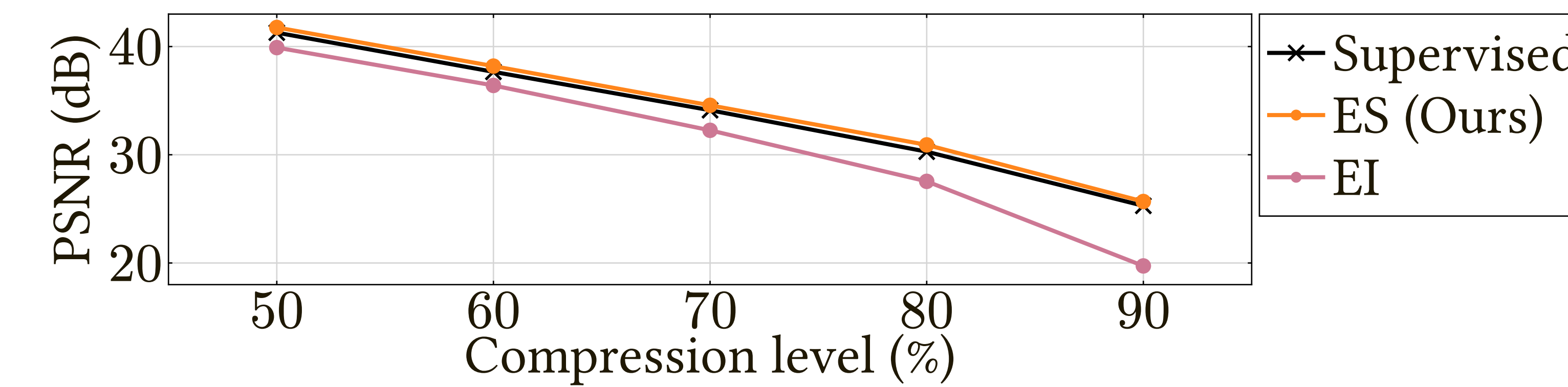


Theorem (Efficient loss evaluation)

If $f(y, A)$ is an equivariant reconstructor then the ES loss reduces to

$$\mathcal{L}_{\text{ES}}(y, A, f) = \mathbb{E}_{y_1, A_1 | y, A} \left\{ \|Af(y_1, A_1) - y\|^2 \right\}$$

Compressive sensing results



Reconstruction performance (PSNR in dB)

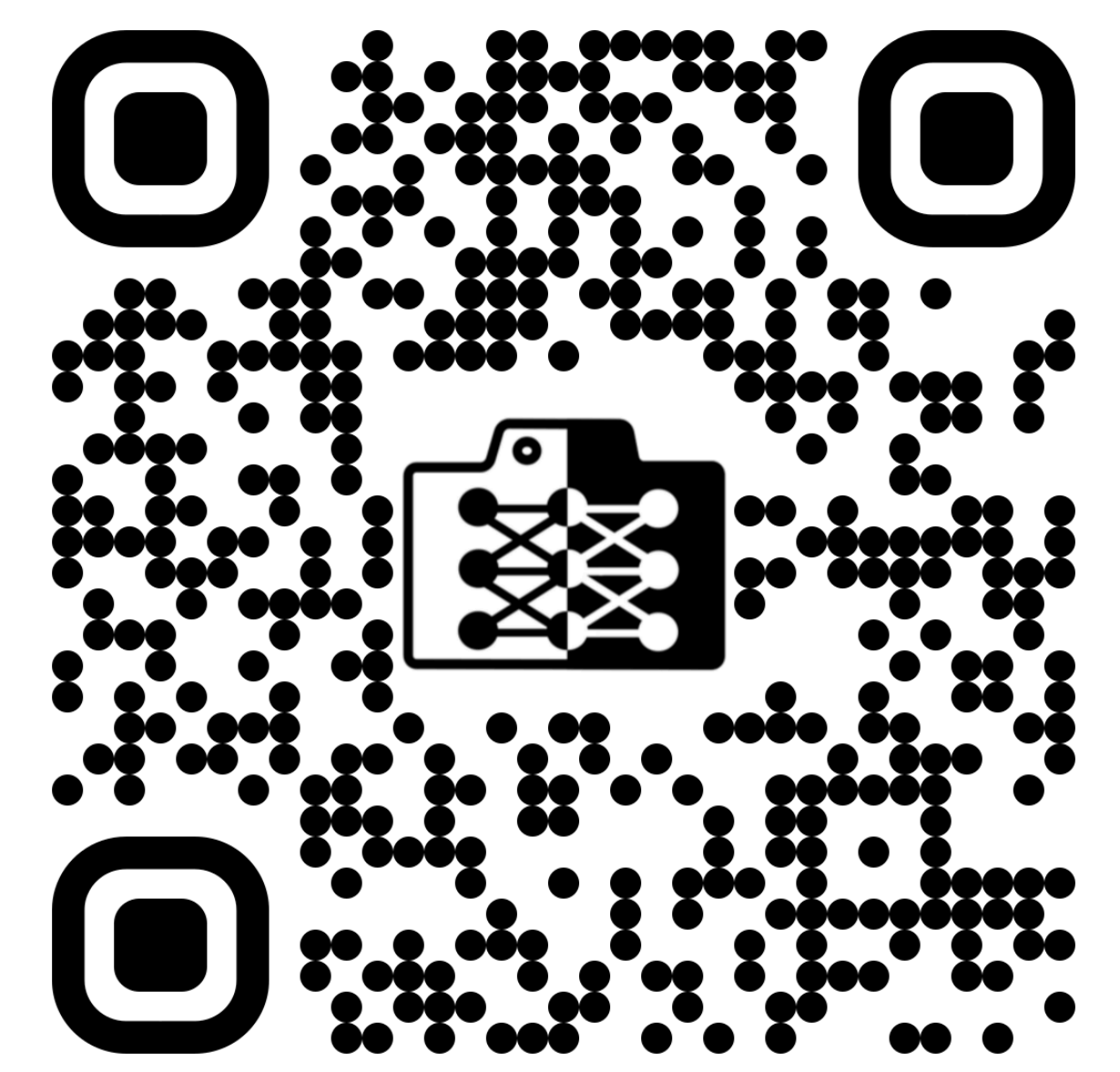
Method	Inpainting 30% rate	MRI ×8 accel.	CT 50 views
Supervised [1]	28.5	28.8	34.0
ES (Ours)	27.5	28.3	32.6
EI [2]	25.9	27.9	28.6

Decision table for the transformations

Operator	Translation	Rotation	Permutation
Compr. sensing	×	✓	✓
Inpainting	✓	✓	✓
Mag. res. imag.	×	✓	✓
Tomography	×	✓	✓
Deblurring	×	×	✓



Equivariant Splitting



DeepInverse

References

- [1] K. H. Jin, M. T. McCann, E. Froustey, and M. Unser, "Deep Convolutional Neural Network for Inverse Problems in Imaging," *IEEE TIP*, 2017.
- [2] D. Chen, J. Tachella, and M. E. Davies, "Robust Equivariant Imaging: A Fully Unsupervised Framework for Learning to Image from Noisy and Partial Measurements," in *CVPR*, 2022.
- [3] C. Millard and M. Chiew, "A Theoretical Framework for Self-Supervised MR Image Reconstruction Using Sub-Sampling via Variable Density Noisier2Noise," *IEEE TCI*, 2023.